Dikaios Amanah

A Secure, Transparent and Ethical SME Investment Platform Based on Musharakah Principles



Software Requirements Specification

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements of Dikaios Amanah, a secure, transparent, and ethically governed web-based platform designed to connect Small and Medium Enterprises (SMEs) with potential investors through a structured investment process.

The purpose of this document is to establish a clear and detailed specification of the system's expected behavior, functional capabilities, non-functional requirements, user interactions, data requirements, system constraints, and testing requirements. The SRS will serve as the primary reference for the design, development, testing, and evaluation of the Dikaios Amanah system.

The system is designed around the Musharakah partnership model, where investment participants share profits and losses according to mutually agreed terms. The proposed platform emphasizes transparency, verification, security, and traceability throughout the investment lifecycle.

1.2 Scope

Dikaios Amanah will provide a web-based environment through which verified business owners can present eligible SME investment opportunities and investors can review and participate in those opportunities.

The system will support three primary user roles:

- Business Owner
- Investor
- Administrator

Business owners will be able to create and manage business profiles, submit required verification information, upload supporting documents, create investment campaigns after approval, provide business progress information, and manage campaign-related information.

Investors will be able to create accounts, complete the required verification process, browse approved investment campaigns, review business and campaign information, participate in simulated investment transactions, and monitor their investment portfolio and history.

Administrators will be responsible for platform-level management activities including user verification, business approval, document review, campaign moderation, risk assessment, user management, and monitoring of system activity.

The system will also provide an audit and trust mechanism for important system events. The proposed architecture includes a Permissioned Trust and Audit Layer intended to improve transparency and provide tamper-evident records of critical activities.

The Minimum Viable Product (MVP) will focus on the core platform workflow required for academic demonstration and evaluation. Real banking integration, live payment processing, production-level regulatory implementation, AI-based fraud detection, AI-powered risk prediction, production Hyperledger Fabric deployment, and automated smart-contract execution are outside the scope of the MVP.

1.3 Intended Audience

This document is intended for the following audiences:

- Project developer and system designer
- Software Engineering course instructor and evaluator
- Software testing and quality assurance personnel
- Project stakeholders
- Future developers who may extend the system

The document provides the requirements and technical specifications needed to understand the expected behavior and structure of the proposed system.

1.4 Product Overview

Dikaios Amanah is intended to function as a technology-enabled platform connecting verified SMEs seeking investment with verified investors seeking structured partnership-based investment opportunities.

The proposed system follows the Musharakah principle rather than interest-based lending. The platform is designed to provide a transparent investment lifecycle beginning with registration and verification and continuing through campaign approval, investment participation, agreement management, business reporting, profit declaration, and profit distribution.

The system architecture proposed in the project consists of a web frontend, backend services, a relational database, secure document storage, authentication and security mechanisms, and a Permissioned Trust and Audit Layer.

The application layer is planned around React for frontend development and Node.js with Express.js for backend services. PostgreSQL will be used for relational data management, while Firebase Storage is intended for secure document storage. Authentication and authorization mechanisms will protect user accounts and role-specific functionality.

Critical events and records will be subject to audit mechanisms so that important activities can be reviewed by authorized users. The trust and audit mechanism is intended to complement the primary application database rather than replace it.

1.5 Definitions and Acronyms

SME Small and Medium Enterprise.

MVP Minimum Viable Product.

KYC Know Your Customer.

RBAC Role-Based Access Control.

JWT JSON Web Token.

API Application Programming Interface.

HTTPS Hypertext Transfer Protocol Secure.

SHA-256 Secure Hash Algorithm 256-bit.

SRS Software Requirements Specification.

UML Unified Modeling Language.

SDLC Software Development Life Cycle.

ERD Entity Relationship Diagram.

1.6 References

The primary project-specific source for this Software Requirements Specification is the Dikaios Amanah Project Proposal. The proposal establishes the project's background, objectives, scope, proposed solution, system architecture, technology stack, feasibility considerations, limitations, and planned development activities.

Additional technical and domain references are listed in the References section of this document.

2. Overall Description

2.1 Product Perspective

Dikaios Amanah is a web-based software platform designed to facilitate structured interaction between verified Small and Medium Enterprises (SMEs), retail investors, and system administrators.

The system follows a layered three-tier architecture consisting of a presentation layer, an application layer, and a data layer. The separation of these layers allows user interaction, application logic, security controls, and persistent data management to be handled independently.

The presentation layer provides the web interface for Business Owners, Investors, and Administrators. The application layer processes requests, enforces business rules, manages authentication and authorization, and coordinates the major platform services. The data layer manages operational records, business information, investment information, and supporting documents.

The proposed system is intended to operate as a technology-enabled intermediary rather than as a financial institution. Investment transactions within the academic MVP will be simulated for demonstration and testing purposes.

2.2 Product Functions

The major functions of Dikaios Amanah are organized according to the responsibilities of the system's primary users.

2.2.1 User Registration and Authentication

The system will provide account registration and authentication mechanisms for users. Authentication will use secure password handling and token-based session management. Access to system resources will be controlled according to the user's assigned role.

2.2.2 Business Owner Functions

Business Owners will be able to:

- Register an account.
- Complete identity and business verification.
- Manage business profile information.
- Submit supporting documents.
- Create investment campaigns after approval.
- Monitor campaign progress.
- Provide business progress information.
- Declare business profits.
- Communicate relevant information to investors.

A business must complete the required verification and administrative review before its investment campaign becomes available to investors.

2.2.3 Investor Functions

Investors will be able to:

- Register an account.

- Complete the required KYC verification process.
- Discover verified businesses and approved campaigns.
- Review business and campaign information.
- Review available risk information.
- Participate in simulated investment transactions.
- Monitor investments.
- View portfolio information.
- Track investment history.
- Monitor declared profits.

2.2.4 Administrator Functions

Administrators will be responsible for maintaining platform integrity and managing operational activities. Administrative functions will include:

- User verification.
- Business approval and rejection.
- Document validation.
- Risk assessment management.
- Campaign moderation.
- User management.
- Audit monitoring.
- Platform activity monitoring.
- Platform reporting.

Administrative actions will be recorded through the system's activity and audit mechanisms.

2.2.5 Investment Campaign Management

The system will allow approved Business Owners to create and manage investment campaigns. Campaign information will be presented to Investors after the required administrative approval.

Investors will be able to review approved campaigns before participating. The campaign lifecycle will support the proposed workflow from campaign publication through investment participation and eventual completion.

2.2.6 Document Management

The system will allow Business Owners to submit supporting documents during the verification process. Documents will be stored using the planned secure cloud storage mechanism.

Cryptographic hashing will be used to support document integrity verification and help identify unauthorized modification of stored document content.

2.2.7 Trust and Audit Functions

The system will maintain records of important platform events through a Permissioned Trust and Audit Layer.

Critical events identified by the project proposal include:

- Business verification.
- Contract approval.
- Investment confirmation.
- Profit declaration.
- Audit records.
- Document integrity verification.

The trust and audit mechanism will complement the primary relational database rather than replace it.

2.2.8 Notification and Activity Functions

The system will provide notifications and activity information to help users remain aware of relevant actions and changes within the platform.

Examples include verification status changes, campaign-related updates, investment-related activities, and administrative actions.

2.3 User Classes and Characteristics

Dikaios Amanah contains three primary user classes.

2.3.1 Business Owner

A Business Owner represents an SME seeking investment through the platform. Business Owners are expected to provide accurate business information and supporting documentation. They will primarily interact with the platform through the Business Owner Dashboard.

2.3.2 Investor

An Investor represents an individual seeking to participate in verified Musharakah-based investment opportunities. Investors will use the platform to review business information, evaluate available opportunities, manage simulated investments, and monitor their portfolios.

2.3.3 Administrator

An Administrator is an authorized platform operator responsible for verification, approval, monitoring, moderation, and other administrative activities. Administrators have access to functions that are restricted from Business Owners and Investors.

2.4 Operating Environment

The proposed system will operate as a web-based application.

The planned software environment consists of:

- React.js for the frontend application.

- Node.js as the backend runtime environment.
- Express.js for backend and RESTful API services.
- PostgreSQL for relational data management.
- Firebase Storage for secure document storage.
- JWT for authentication.
- bcrypt for password hashing.
- SHA-256 for document integrity verification.
- Git and GitHub for version control.

Users will access the system through modern web browsers using an internet connection. Communication between the frontend and backend services will use HTTPS.

2.5 Design Constraints

The following constraints are derived from the defined project scope and academic MVP objectives:

- The project must remain within the defined academic MVP scope.
- Investment transactions will be simulated rather than connected to real financial payment systems.
- Live banking integration is excluded from the MVP.
- Real payment gateway integration is excluded from the MVP.
- Production-level BSEC regulatory compliance implementation is excluded from the MVP.
- AI-based fraud detection is excluded from the MVP.
- AI-powered risk prediction is excluded from the MVP.
- Production Hyperledger Fabric deployment is excluded from the MVP.

- Automated smart contract execution is excluded from the MVP.
- Native Android and iOS applications are excluded from the MVP.

These limitations are intended to keep the system technically achievable within the project's academic development scope.

2.6 Assumptions and Dependencies

The system development is based on several assumptions established in the project proposal:

- Users have basic digital literacy.
- Users have access to an internet connection.
- Businesses provide authentic documentation.
- Administrators perform verification activities responsibly.
- The system is initially developed for academic demonstration purposes.
- Investment transactions used during testing are simulated.

The system also depends on the availability and proper operation of its planned software and infrastructure components, including the relational database, document storage service, authentication mechanisms, and deployment environment.

3. Functional Requirements

3.1 Authentication and Account Management

3.1.1 FR-AUTH-01: User Registration

The system shall allow a new user to register an account by providing the required personal and account information.

3.1.2 FR-AUTH-02: User Login

The system shall authenticate registered users using their valid login credentials.

3.1.3 FR-AUTH-03: Role-Based Access

The system shall assign users an appropriate system role and restrict access to functions according to the user's authorized role.

The primary roles shall be:

- Business Owner
- Investor
- Administrator

3.1.4 FR-AUTH-04: Secure Password Storage

The system shall store user passwords using a secure password-hashing mechanism rather than storing plaintext passwords.

3.1.5 FR-AUTH-05: Session Authentication

The system shall maintain authenticated user sessions using a secure token-based authentication mechanism.

3.1.6 FR-AUTH-06: Account Management

The system shall allow authenticated users to view and manage their permitted account and profile information.

3.2 Business Owner Requirements

3.2.1 FR-BUS-01: Business Profile Creation

The system shall allow an authenticated Business Owner to create a business profile containing the required business information.

3.2.2 FR-BUS-02: Business Profile Management

The system shall allow a Business Owner to update permitted business profile information.

3.2.3 FR-BUS-03: Verification Submission

The system shall allow a Business Owner to submit the required business verification information for administrative review.

3.2.4 FR-BUS-04: Document Submission

The system shall allow a Business Owner to upload supporting documents required for the verification process.

3.2.5 FR-BUS-05: Verification Status

The system shall allow a Business Owner to view the current status of the business verification process.

3.2.6 FR-BUS-06: Campaign Creation

The system shall allow an approved Business Owner to create an investment campaign containing the required campaign information.

3.2.7 FR-BUS-07: Campaign Management

The system shall allow an authorized Business Owner to view and manage permitted information associated with their campaigns.

3.2.8 FR-BUS-08: Campaign Progress

The system shall allow a Business Owner to monitor the funding progress of an approved campaign.

3.2.9 FR-BUS-09: Business Progress Reporting

The system shall allow a Business Owner to submit business progress information relevant to an active campaign.

3.2.10 FR-BUS-10: Profit Declaration

The system shall allow an authorized Business Owner to record declared business profit information associated with an investment campaign.

3.3 Investor Requirements

3.3.1 FR-INV-01: Investor Verification

The system shall allow an Investor to submit the required information for the investor verification process.

3.3.2 FR-INV-02: Investor Dashboard

The system shall provide an authenticated Investor with a dashboard containing relevant investment and account information.

3.3.3 FR-INV-03: Campaign Discovery

The system shall allow verified Investors to browse approved investment campaigns.

3.3.4 FR-INV-04: Campaign Information

The system shall allow an Investor to view relevant information about an approved investment campaign before participating.

3.3.5 FR-INV-05: Business Information

The system shall allow an Investor to view permitted information about the business associated with an approved investment campaign.

3.3.6 FR-INV-06: Risk Information

The system shall present available risk assessment information associated with an investment opportunity.

3.3.7 FR-INV-07: Simulated Investment

The system shall allow an eligible Investor to participate in a simulated investment transaction during the MVP.

3.3.8 FR-INV-08: Investment Confirmation

The system shall provide confirmation after a simulated investment has been successfully recorded.

3.3.9 FR-INV-09: Portfolio Management

The system shall allow an Investor to view their investment portfolio.

3.3.10 FR-INV-10: Investment History

The system shall allow an Investor to view the history of their recorded investment activities.

3.3.11 FR-INV-11: Profit Information

The system shall allow an Investor to view declared profit information associated with their investments.

3.4 Administrator Requirements

3.4.1 FR-ADMIN-01: Administrator Dashboard

The system shall provide an authenticated Administrator with an administrative dashboard containing relevant platform information.

3.4.2 FR-ADMIN-02: User Management

The system shall allow authorized Administrators to view and manage user accounts according to their administrative permissions.

3.4.3 FR-ADMIN-03: Business Verification Review

The system shall allow an Administrator to review submitted business verification information.

3.4.4 FR-ADMIN-04: Business Approval

The system shall allow an Administrator to approve or reject a business verification request.

3.4.5 FR-ADMIN-05: Document Review

The system shall allow an Administrator to review documents submitted as part of the verification process.

3.4.6 FR-ADMIN-06: Risk Assessment Management

The system shall allow an authorized Administrator to manage the risk assessment information associated with eligible businesses or campaigns.

3.4.7 FR-ADMIN-07: Campaign Moderation

The system shall allow an Administrator to review and moderate submitted investment campaigns.

3.4.8 FR-ADMIN-08: Campaign Approval

The system shall allow an Administrator to approve or reject a submitted investment campaign.

3.4.9 FR-ADMIN-09: Platform Activity Monitoring

The system shall allow an Administrator to monitor relevant platform activities.

3.4.10 FR-ADMIN-10: Platform Reporting

The system shall provide authorized Administrators with access to relevant platform reports.

3.5 Investment Campaign Management

3.5.1 FR-CAM-01: Campaign Submission

The system shall allow an approved Business Owner to submit an investment campaign for administrative review.

3.5.2 FR-CAM-02: Campaign Approval Status

The system shall maintain the approval status of each submitted campaign.

3.5.3 FR-CAM-03: Campaign Publication

The system shall make an approved campaign available to eligible Investors.

3.5.4 FR-CAM-04: Campaign Information Display

The system shall display the relevant campaign information required for an Investor to review an investment opportunity.

3.5.5 FR-CAM-05: Campaign Funding Progress

The system shall maintain and display the recorded funding progress of an active campaign.

3.5.6 FR-CAM-06: Campaign Lifecycle

The system shall maintain the status of a campaign throughout its defined lifecycle, including relevant approval, active, and completed states.

3.6 Document Management Requirements

3.6.1 FR-DOC-01: Document Upload

The system shall allow authorized users to upload documents required for verification and other permitted platform activities.

3.6.2 FR-DOC-02: Secure Document Storage

The system shall store uploaded documents using the designated secure document storage service.

3.6.3 FR-DOC-03: Document Integrity Hash

The system shall generate a cryptographic hash for applicable uploaded documents to support document integrity verification.

3.6.4 FR-DOC-04: Document Verification

The system shall allow authorized Administrators to review the integrity information associated with submitted documents.

3.7 Notification and Activity Requirements

3.7.1 FR-NOT-01: User Notifications

The system shall provide notifications for relevant events affecting authorized users.

3.7.2 FR-NOT-02: Verification Notification

The system shall notify relevant users when the status of a verification request changes.

3.7.3 FR-NOT-03: Campaign Notification

The system shall provide relevant campaign-related activity notifications to authorized users.

3.7.4 FR-NOT-04: Activity History

The system shall maintain a history of relevant user and platform activities.

3.8 Trust and Audit Requirements

3.8.1 FR-AUD-01: Critical Event Recording

The system shall record defined critical system events through the Permissioned Trust and Audit Layer.

3.8.2 FR-AUD-02: Business Verification Audit

The system shall record relevant business verification events in the audit record.

3.8.3 FR-AUD-03: Contract Approval Audit

The system shall record relevant contract approval events in the audit record.

3.8.4 FR-AUD-04: Investment Confirmation Audit

The system shall record relevant investment confirmation events in the audit record.

3.8.5 FR-AUD-05: Profit Declaration Audit

The system shall record relevant profit declaration events in the audit record.

3.8.6 FR-AUD-06: Audit Record Access

The system shall allow authorized users to view permitted audit and activity information.

3.8.7 FR-AUD-07: Audit Integrity

The system shall provide a mechanism for detecting unauthorized alteration of applicable audit records.

3.9 Investment Lifecycle Requirements

3.9.1 FR-LIFE-01: Investment Participation

The system shall associate a recorded simulated investment with the corresponding Investor and approved campaign.

3.9.2 FR-LIFE-02: Investment Record

The system shall maintain the information necessary to identify a recorded investment, its associated campaign, and its participating Investor.

3.9.3 FR-LIFE-03: Profit Record

The system shall maintain declared profit information associated with eligible investment activities.

3.9.4 FR-LIFE-04: Investment Status

The system shall maintain the status of recorded investment activities throughout the supported investment lifecycle.

4. Non-Functional Requirements

4.1 Security Requirements

4.1.1 NFR-SEC-01: Authentication Security

The system shall require users to authenticate before accessing protected system functions.

4.1.2 NFR-SEC-02: Role-Based Authorization

The system shall restrict access to protected resources according to the authenticated user's assigned role.

4.1.3 NFR-SEC-03: Password Protection

The system shall not store user passwords in plaintext. Passwords shall be protected using a secure password-hashing mechanism.

4.1.4 NFR-SEC-04: Secure Communication

The deployed system shall use HTTPS for communication between clients and server-side services.

4.1.5 NFR-SEC-05: Document Integrity

The system shall use cryptographic hashing to support verification of the integrity of applicable uploaded documents.

4.1.6 NFR-SEC-06: Audit Protection

The system shall provide mechanisms to detect unauthorized alteration of applicable audit records.

4.1.7 NFR-SEC-07: Sensitive Data Protection

The system shall restrict access to sensitive business, investor, verification, and administrative information to authorized users.

4.2 Performance Requirements

4.2.1 NFR-PERF-01: Normal Request Response

Under the expected workload of the academic MVP, normal user requests should receive a response within an acceptable period that does not unnecessarily interrupt the user's workflow.

4.2.2 NFR-PERF-02: Dashboard Loading

The system should load normal user dashboard information without excessive delay under the expected academic demonstration workload.

4.2.3 NFR-PERF-03: Database Operations

The system shall use appropriate database queries and indexing strategies to support efficient retrieval of frequently accessed application data.

4.2.4 NFR-PERF-04: Document Operations

Document upload and retrieval operations should provide appropriate feedback to users while the operation is being processed.

4.3 Usability Requirements

4.3.1 NFR-USE-01: Consistent Interface

The system shall provide a consistent user interface across the major platform modules.

4.3.2 NFR-USE-02: Role-Specific Navigation

The system shall provide navigation appropriate to the authenticated user's role.

4.3.3 NFR-USE-03: Form Validation

The system shall provide appropriate validation and feedback when users submit forms.

4.3.4 NFR-USE-04: Error Feedback

The system shall provide understandable error messages when an operation cannot be completed.

4.3.5 NFR-USE-05: Responsive Interface

The web interface should support common desktop, tablet, and mobile screen sizes.

4.3.6 NFR-USE-06: Status Visibility

The system shall clearly communicate important statuses such as verification, campaign approval, investment, and other relevant workflow states.

4.4 Reliability Requirements

4.4.1 NFR-REL-01: Data Consistency

The system shall maintain consistency between related application records.

4.4.2 NFR-REL-02: Transaction Integrity

The system shall ensure that supported investment and other critical operations either complete successfully or do not leave inconsistent application records.

4.4.3 NFR-REL-03: Error Handling

The system shall handle expected application errors without exposing sensitive implementation details to users.

4.4.4 NFR-REL-04: Data Persistence

Successfully stored application records shall remain available after normal application sessions have ended.

4.5 Maintainability Requirements

4.5.1 NFR-MNT-01: Modular Architecture

The system shall maintain a modular separation between frontend, backend, database, and supporting services.

4.5.2 NFR-MNT-02: Source Code Organization

The source code shall follow a structured organization that allows individual components to be modified without unnecessary changes to unrelated components.

4.5.3 NFR-MNT-03: Version Control

The project source code and documentation shall be maintained using version control.

4.5.4 NFR-MNT-04: Documentation

Major system components and interfaces shall be documented sufficiently to support future maintenance and development.

4.5.5 NFR-MNT-05: Configuration Separation

Environment-specific configuration and sensitive credentials shall be kept separate from the application source code.

4.6 Scalability Requirements

4.6.1 NFR-SCL-01: Modular Expansion

The architecture should allow additional platform features to be added without requiring a complete redesign of the existing system.

4.6.2 NFR-SCL-02: User Growth

The system architecture should allow the number of users and application records to increase beyond the initial academic demonstration workload.

4.6.3 NFR-SCL-03: Service Expansion

The system should allow future integration with additional external services where required.

4.7 Availability Requirements

4.7.1 NFR-AVL-01: Demonstration Availability

The deployed MVP should remain accessible during scheduled project demonstrations and evaluation periods.

4.7.2 NFR-AVL-02: Service Recovery

The system should provide a practical procedure for restoring the application after an expected service or deployment failure.

4.7.3 NFR-AVL-03: Dependency Availability

The system shall communicate appropriate errors when a required external service is temporarily unavailable.

4.8 Compatibility Requirements

4.8.1 NFR-CMP-01: Browser Compatibility

The web application should operate correctly on current versions of commonly used modern web browsers.

4.8.2 NFR-CMP-02: Web-Based Access

The system shall provide access through a standard web browser without requiring a dedicated desktop application.

4.9 Auditability Requirements

4.9.1 NFR-AUD-01: Critical Event Traceability

Critical system activities shall provide sufficient information to identify the relevant event and its associated system entities.

4.9.2 NFR-AUD-02: Audit Record Integrity

The audit mechanism shall support detection of unauthorized modification of applicable records.

4.9.3 NFR-AUD-03: Authorized Audit Access

Audit information shall only be accessible to users with appropriate permissions.

5. External Interface Requirements

5.1 User Interface Requirements

5.1.1 UI-01: Web-Based Interface

The system shall provide a web-based graphical user interface through which Business Owners, Investors, and Administrators can access functions appropriate to their roles.

5.1.2 UI-02: Authentication Interface

The system shall provide user interfaces for registration, login, and account-related operations.

5.1.3 UI-03: Business Owner Interface

The Business Owner interface shall provide access to functions including:

- Business profile management.
- Verification submission.
- Document submission.

- Verification status.
- Campaign creation and management.
- Campaign progress.
- Business progress reporting.
- Profit declaration.

5.1.4 UI-04: Investor Interface

The Investor interface shall provide access to functions including:

- Investor dashboard.
- Campaign discovery.
- Business and campaign information.
- Risk information.
- Simulated investment.
- Investment confirmation.
- Portfolio management.
- Investment history.
- Profit information.

5.1.5 UI-05: Administrator Interface

The Administrator interface shall provide access to functions including:

- User management.
- Business verification.
- Document review.
- Risk assessment management.

- Campaign moderation.
- Campaign approval.
- Platform activity monitoring.
- Platform reporting.
- Audit information.

5.1.6 UI-06: Navigation

The interface shall provide navigation appropriate to the authenticated user's role and shall prevent users from accessing functions outside their authorized permissions.

5.1.7 UI-07: Form Validation

Input forms shall provide appropriate validation and feedback for invalid or incomplete information.

5.1.8 UI-08: Status Indicators

The interface shall clearly display relevant workflow states, including verification, approval, campaign, investment, and other applicable statuses.

5.1.9 UI-09: Feedback Messages

The system shall provide appropriate success, warning, and error messages after user actions.

5.1.10 UI-10: Responsive Design

The interface should adapt to commonly used desktop, tablet, and mobile screen sizes.

5.2 Software Interfaces

5.2.1 SI-01: Frontend and Backend Interface

The frontend application shall communicate with backend services through defined application programming interfaces (APIs).

5.2.2 SI-02: RESTful API

The backend shall provide RESTful API services for communication between the web frontend and application services.

5.2.3 SI-03: Authentication Interface

The authentication mechanism shall provide the frontend with an authenticated session or token that can be used when accessing protected backend resources.

5.2.4 SI-04: Database Interface

The backend shall communicate with the PostgreSQL database through an appropriate database access mechanism.

The database interface shall support operations required for managing users, businesses, campaigns, investments, verification information, activity records, and other application data.

5.2.5 SI-05: Document Storage Interface

The backend shall communicate with the designated secure document storage service for storing and retrieving permitted uploaded documents.

5.2.6 SI-06: Trust and Audit Interface

The application services shall communicate with the Permissioned Trust and Audit Layer when critical events require recording or verification.

5.2.7 SI-07: External Service Failure Handling

The system shall provide appropriate error handling when a required external service is unavailable or fails to respond.

5.3 Communication Interfaces

5.3.1 CI-01: HTTPS Communication

The deployed application shall use HTTPS for communication between clients and server-side services.

5.3.2 CI-02: Client-Server Communication

The frontend shall communicate with the backend through HTTP-based API requests.

5.3.3 CI-03: API Data Format

The application API should use JSON as the primary format for structured request and response data.

5.3.4 CI-04: Authentication Headers

Protected API requests shall include the required authentication information so that the backend can verify the identity and authorization of the requesting user.

5.3.5 CI-05: Error Responses

The backend shall return structured error responses that allow the frontend to provide appropriate feedback to users.

5.4 Hardware Interfaces

Dikaios Amanah does not require specialized hardware for normal operation.

Users will access the system using standard computing devices such as desktop computers, laptops, tablets, or smartphones with compatible web browsers and network connectivity.

The academic MVP does not require dedicated hardware interfaces or embedded devices.

5.5 Storage Interfaces

5.5.1 ST-01: Relational Data Storage

The system shall use PostgreSQL as the primary relational database for persistent application data.

5.5.2 ST-02: Document Storage

Uploaded documents shall be stored using the designated secure cloud storage mechanism.

5.5.3 ST-03: Document Integrity Information

The system shall maintain the required cryptographic integrity information associated with applicable uploaded documents.

5.5.4 ST-04: Audit Data

The system shall maintain information necessary to associate audit events with the relevant users, entities, and system activities.

5.6 Interface Security Requirements

5.6.1 IS-01: Protected Endpoints

Backend endpoints containing protected functionality shall require appropriate authentication and authorization.

5.6.2 IS-02: Role Verification

The backend shall verify the user's authorized role before executing role-restricted operations.

5.6.3 IS-03: Secure Data Transmission

Sensitive information transmitted between the client and deployed backend shall be protected through HTTPS.

5.6.4 IS-04: Input Validation

The backend shall validate incoming data before processing or storing it.

5.6.5 IS-05: Unauthorized Access Handling

The system shall reject requests that do not satisfy the required authentication or authorization conditions.

6. System Design

6.1 System Architecture

Dikaios Amanah will follow a layered web application architecture designed to separate user interaction, application processing, persistent data management, document storage, and trust and audit functions.

The major components of the proposed architecture are:

- Presentation Layer
- Application and API Layer
- Database Layer
- Document Storage
- Authentication and Security Services
- Permissioned Trust and Audit Layer

The overall architecture is intended to provide separation of responsibilities while allowing the individual components to communicate through defined interfaces.

The high-level architecture can be represented as follows:

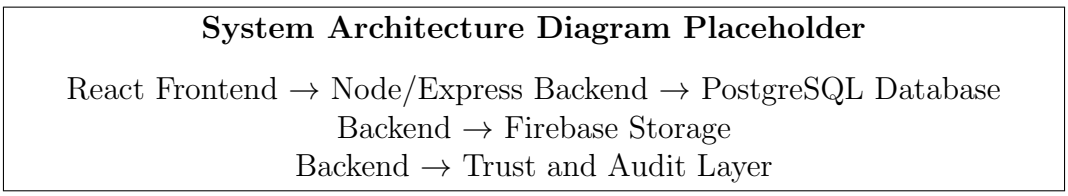


Figure 1: High-Level System Architecture

The final architecture diagram will be added to the SRS after the individual system components and data relationships have been finalized.

6.2 Presentation Layer

The Presentation Layer provides the user-facing web interface of Dikaios Amanah.

The planned frontend will use React and will provide role-specific interfaces for:

- Business Owners
- Investors
- Administrators

The Presentation Layer will be responsible for displaying information, collecting user input, performing client-side validation where appropriate, and communicating with back-end services through defined APIs.

Business logic and security-sensitive authorization decisions shall not depend solely on client-side controls.

6.3 Application and API Layer

The Application Layer will provide the primary business logic of the system.

The planned backend will use Node.js with Express.js to provide RESTful API services.

The Application Layer will be responsible for:

- Processing client requests.
- Enforcing application business rules.
- Authenticating users.
- Authorizing role-specific operations.
- Managing business workflows.
- Processing investment-related operations.
- Coordinating database operations.

- Coordinating document storage operations.
- Recording critical audit events.

The backend will act as the central coordination layer between the frontend and the system's persistent and supporting services.

6.4 Database Layer

PostgreSQL will be used as the primary relational database.

The database will maintain persistent application information including:

- User accounts.
- Business information.
- Investor information.
- Verification records.
- Investment campaigns.
- Investment records.
- Profit information.
- Activity records.
- References to uploaded documents.

The database design will maintain appropriate relationships and integrity constraints between related entities.

The detailed database structure and Entity Relationship Diagram will be specified in the Data Requirements section.

6.5 Document Storage Component

Business verification and related processes may require supporting documents.

The proposed system will use Firebase Storage as the designated document storage mechanism.

The application database will store the relevant metadata and references required to identify uploaded documents, while the actual document files will be maintained by the designated storage service.

The system will generate cryptographic integrity information for applicable documents so that authorized users can verify whether the document content has been altered.

6.6 Authentication and Authorization

Authentication will establish the identity of a user before protected system resources are accessed.

The proposed system will use token-based authentication, with JWT planned for authenticated communication between the frontend and backend.

Authorization will be implemented through Role-Based Access Control (RBAC).

The three primary roles are:

1. Business Owner
2. Investor
3. Administrator

Role verification shall occur at the backend level before protected operations are executed.

This prevents users from gaining access to restricted operations simply by manipulating the frontend interface.

6.7 Permissioned Trust and Audit Layer

Dikaios Amanah includes a Permissioned Trust and Audit Layer as an additional mechanism for improving transparency and traceability.

The purpose of this layer is not to replace the primary PostgreSQL database. Instead, it is intended to maintain tamper-evident records of selected critical events.

Events that may require audit recording include:

- Business verification.
- Contract approval.

- Investment confirmation.
- Profit declaration.
- Document integrity verification.
- Other critical administrative activities.

The trust layer is intended to provide an additional level of confidence that important historical events have not been silently altered.

The MVP will implement the trust and audit concept at an appropriate level for academic demonstration. Production-scale distributed ledger deployment is outside the defined MVP scope.

6.8 System Component Interaction

The major system components will interact through the following general workflow:

1. A user interacts with the React-based frontend.
2. The frontend sends a request to the backend API.
3. The backend authenticates and authorizes the request.
4. The backend applies the appropriate business rules.
5. Required information is retrieved from or stored in PostgreSQL.
6. If documents are involved, the backend communicates with the designated document storage service.
7. If the operation is a critical auditable event, the appropriate audit information is recorded.
8. The backend returns the result to the frontend.
9. The frontend presents the result to the user.

6.9 Deployment Architecture

The planned deployment architecture will separate the public frontend, backend application services, and persistent data services.

The frontend may be deployed using a platform such as Vercel, while the backend may be deployed on a suitable cloud server or VPS. PostgreSQL and document storage will be provided through appropriate persistent cloud services.

A conceptual deployment structure is:

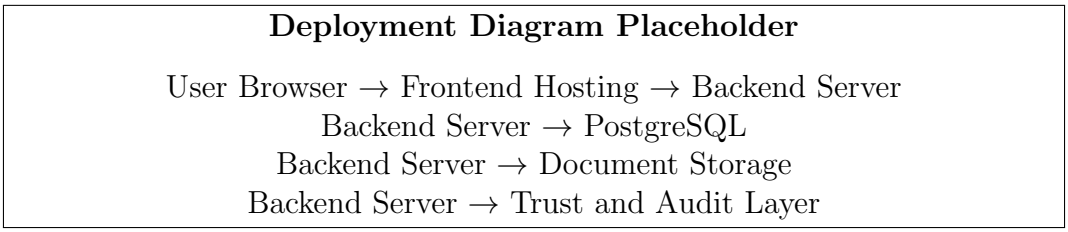


Figure 2: Conceptual Deployment Architecture

The final deployment diagram will be produced after the detailed deployment configuration has been finalized.

6.10 Design Principles

The system design will follow the following principles:

- Separation of frontend and backend responsibilities.
- Separation of application data and document storage.
- Backend-controlled authorization.
- Secure handling of authentication credentials.
- Clear separation between operational data and audit records.
- Modular design to support future expansion.
- Traceability between requirements, implementation, and testing.

7. Use Cases

7.1 Overview

The Use Case Analysis identifies the major functional interactions between the Dikaios Amanah system and its three primary stakeholder groups: Business Owner, Retail Investor, and System Administrator.

The use cases are derived from the functional scope of the proposed system, which includes user authentication, business verification, document management, investment campaign management, investment tracking, portfolio management, profit declaration, notifications, activity logging, and the Permissioned Trust and Audit Layer. These functions support the proposed objective of connecting verified businesses with verified retail investors through a transparent Musharakah-based investment process.

7.2 Primary Actors

The Dikaios Amanah system has three primary actors.

- **Business Owner:** Registers a business, provides business information, uploads verification documents, submits verification requests, creates investment campaigns, monitors campaigns, and declares business profits.
- **Retail Investor:** Registers and completes the required verification process, browses verified businesses and investment campaigns, participates in investment opportunities, and monitors investment and portfolio information.
- **System Administrator:** Manages users, reviews business verification requests, validates submitted information, approves or rejects businesses, monitors system activities, and reviews audit records.

7.3 Use Case Diagram

The Use Case Diagram presents the major functional interactions between the three primary actors and the Dikaios Amanah system.

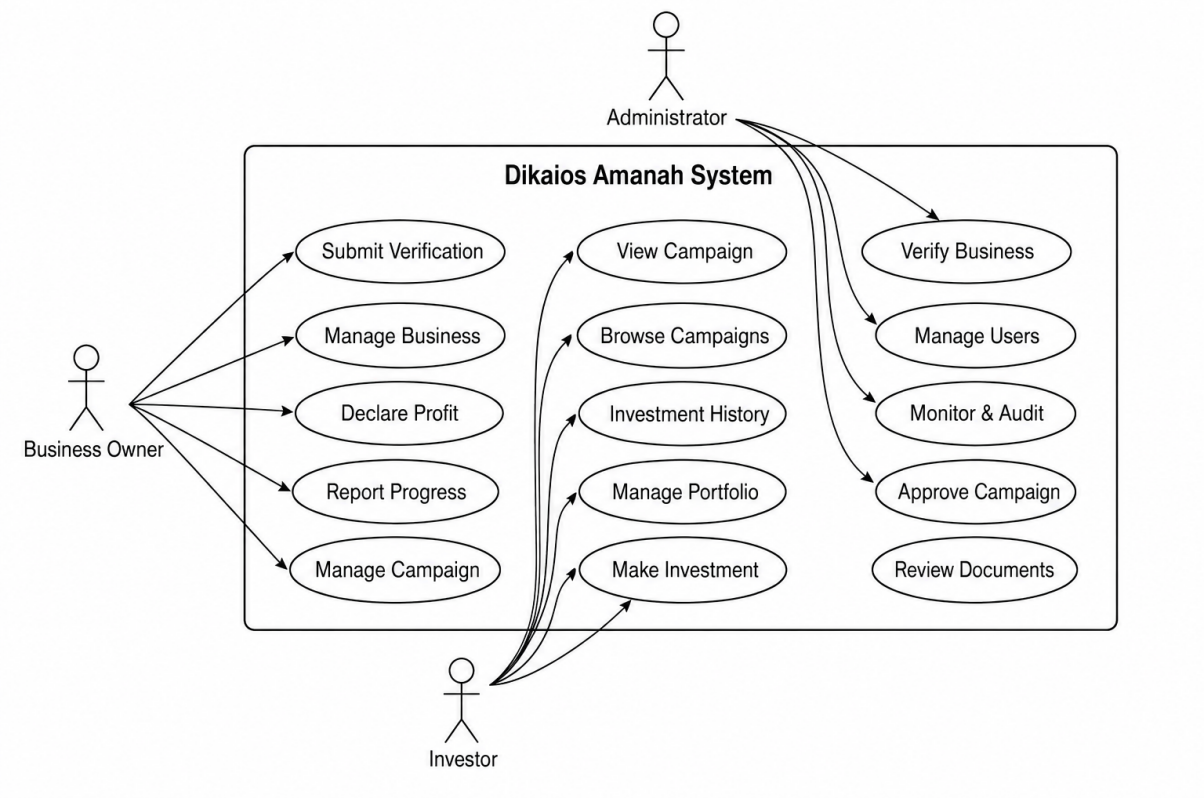


Figure 3: Dikaios Amanah Use Case Diagram

7.4 Business Owner Use Cases

The Business Owner interacts with the system primarily to establish a verified business profile and obtain investment through Musharakah-based campaigns.

7.4.1 Register and Authenticate

The Business Owner creates an account and authenticates using the secure authentication mechanism provided by the system.

Precondition: The user does not already have an active account.

Main Flow:

1. The Business Owner provides registration information.
2. The system validates the submitted information.
3. The system creates the user account.
4. The Business Owner logs into the system.

5. The system authenticates the credentials and provides access to the Business Owner dashboard.

Postcondition: The Business Owner is authenticated and can access authorized system functions.

7.4.2 Submit Business Verification

The Business Owner submits business information and supporting documents for administrative verification.

Precondition: The Business Owner is authenticated.

Main Flow:

1. The Business Owner enters business information.
2. The Business Owner uploads the required documents.
3. The system validates the submitted information.
4. The system stores the verification record and documents.
5. The system notifies the Administrator.

Postcondition: The business verification request is available for Administrator review.

7.4.3 Create Investment Campaign

A verified Business Owner creates an investment campaign containing the required business and funding information.

Precondition: The Business Owner and business have been verified.

Main Flow:

1. The Business Owner selects the campaign creation function.
2. The Business Owner provides campaign title and description.
3. The Business Owner specifies the funding goal and related information.
4. The system validates the information.

5. The system creates the campaign.

Postcondition: The campaign is stored and can be made available to eligible investors.

7.4.4 Monitor Investment Campaign

The Business Owner monitors campaign status and funding progress.

Main Flow:

1. The Business Owner opens the campaign dashboard.
2. The system retrieves campaign information.
3. The system displays current funding and campaign status.

7.4.5 Declare Profit

The Business Owner submits profit information associated with an investment campaign.

Precondition: An applicable investment campaign exists.

Main Flow:

1. The Business Owner selects the profit declaration function.
2. The Business Owner enters the profit amount and reporting period.
3. The system validates the submitted information.
4. The system stores the profit declaration.
5. The event is recorded in the audit layer.

Postcondition: The profit declaration is stored and available for authorized processing and monitoring.

7.5 Retail Investor Use Cases

Retail Investors interact with the system to discover verified investment opportunities, participate in campaigns, and monitor their investments.

7.5.1 Register and Complete Investor Verification

The Retail Investor creates an account and provides the required information for verification.

Main Flow:

1. The Investor registers an account.
2. The system validates the submitted information.
3. The Investor provides required verification information.
4. The system stores the verification information.
5. The Administrator reviews the submitted information.

Postcondition: The Investor account is available according to its verification status.

7.5.2 Browse Verified Businesses

The Investor searches and views verified businesses available on the platform.

Main Flow:

1. The Investor opens the investment marketplace.
2. The system retrieves eligible verified businesses.
3. The system displays business profiles and available campaigns.
4. The Investor selects a business or campaign for further review.

7.5.3 Review Investment Campaign

The Investor reviews the information associated with an investment opportunity before making an investment decision.

The system provides relevant campaign information, business information, funding progress, and available investment details.

7.5.4 Participate in Investment Campaign

The Investor participates in an available Musharakah-based investment campaign.

Precondition: The Investor is authenticated and eligible to participate.

Main Flow:

1. The Investor selects an available campaign.
2. The Investor enters the intended investment amount.
3. The system validates the investment information.
4. The system records the investment.
5. The investment status is updated.
6. The event is recorded in the audit layer.

Postcondition: The investment appears in the Investor's portfolio.

7.5.5 Monitor Investment Portfolio

The Investor monitors investment status and portfolio information.

1. The Investor opens the portfolio dashboard.
2. The system retrieves investment records.
3. The system displays investment status and relevant portfolio information.

7.6 Administrator Use Cases

The Administrator is responsible for verification, approval, monitoring, and administrative control of the platform.

7.6.1 Review Business Verification

The Administrator reviews submitted business verification requests.

Main Flow:

1. The Administrator opens the pending verification list.

2. The system retrieves the submitted verification information.
3. The Administrator reviews business information and documents.
4. The Administrator approves or rejects the verification request.
5. The system updates the verification status.
6. The system records the decision in the audit layer.
7. The Business Owner is notified of the result.

7.6.2 Manage Users

The Administrator manages authorized users and their system status.

Possible administrative actions include reviewing user information, updating account status, and managing access according to the user's role.

7.6.3 Monitor System Activity

The Administrator reviews system activity and important operational events.

The system provides activity records generated by important actions such as verification decisions, investment activities, and other critical operations.

7.6.4 Review Audit Records

The Administrator reviews audit records generated by the Permissioned Trust and Audit Layer.

Audit records provide tamper-evident evidence of important system events and support accountability and transparency.

7.7 Use Case Summary

Table 1: Dikaios Amanah Use Case Summary

Actor	Use Case	Description
Business Owner	Register and Authenticate	Creates an account and securely accesses the platform.
Business Owner	Submit Business Verification	Submits business information and supporting documents for verification.
Business Owner	Create Investment Campaign	Creates a Musharakah-based investment campaign.
Business Owner	Monitor Campaign	Monitors campaign status and funding progress.
Business Owner	Declare Profit	Submits profit information for an investment campaign.
Retail Investor	Register and Verify	Creates an account and completes required verification.
Retail Investor	Browse Businesses	Views verified businesses and available opportunities.
Retail Investor	Review Campaign	Examines investment campaign information before investing.
Retail Investor	Participate in Investment	Records participation in an eligible investment campaign.
Retail Investor	Monitor Portfolio	Views investment status and portfolio information.
Administrator	Review Verification	Reviews business information and supporting documents.
Administrator	Manage Users	Manages user accounts and access status.
Administrator	Monitor Activity	Reviews important platform activities.
Administrator	Review Audit Records	Reviews tamper-evident records of critical system events.

7.8 Business Verification Sequence

The sequence diagram describes the interaction between the Business Owner, Administrator, web interface, backend API, verification service, database, audit service, and notification service during business verification.

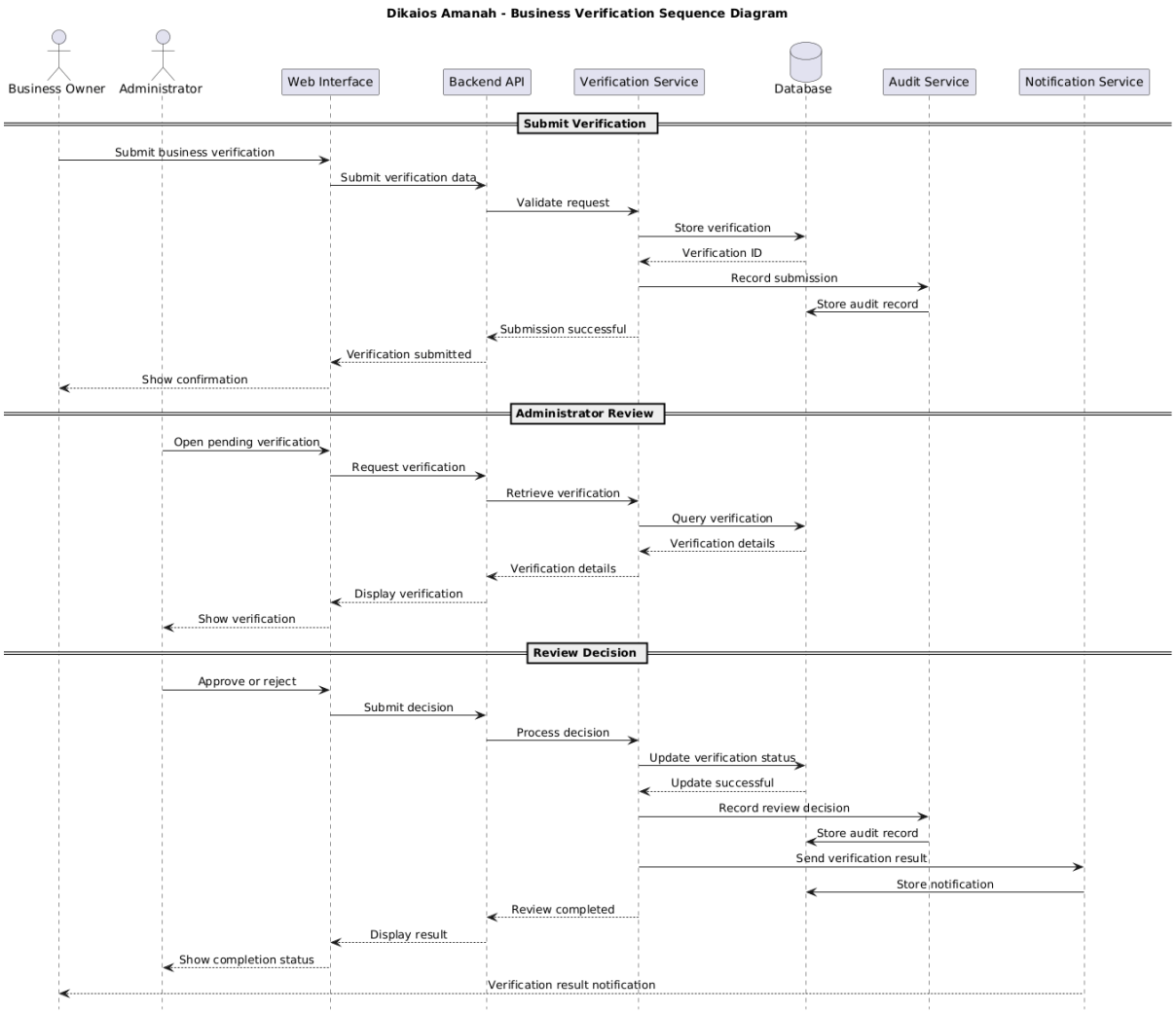


Figure 4: Business Verification Sequence Diagram

7.9 Business Verification Activity

The activity diagram illustrates the workflow beginning with business information submission and continuing through validation, administrative review, approval or rejection, and notification.

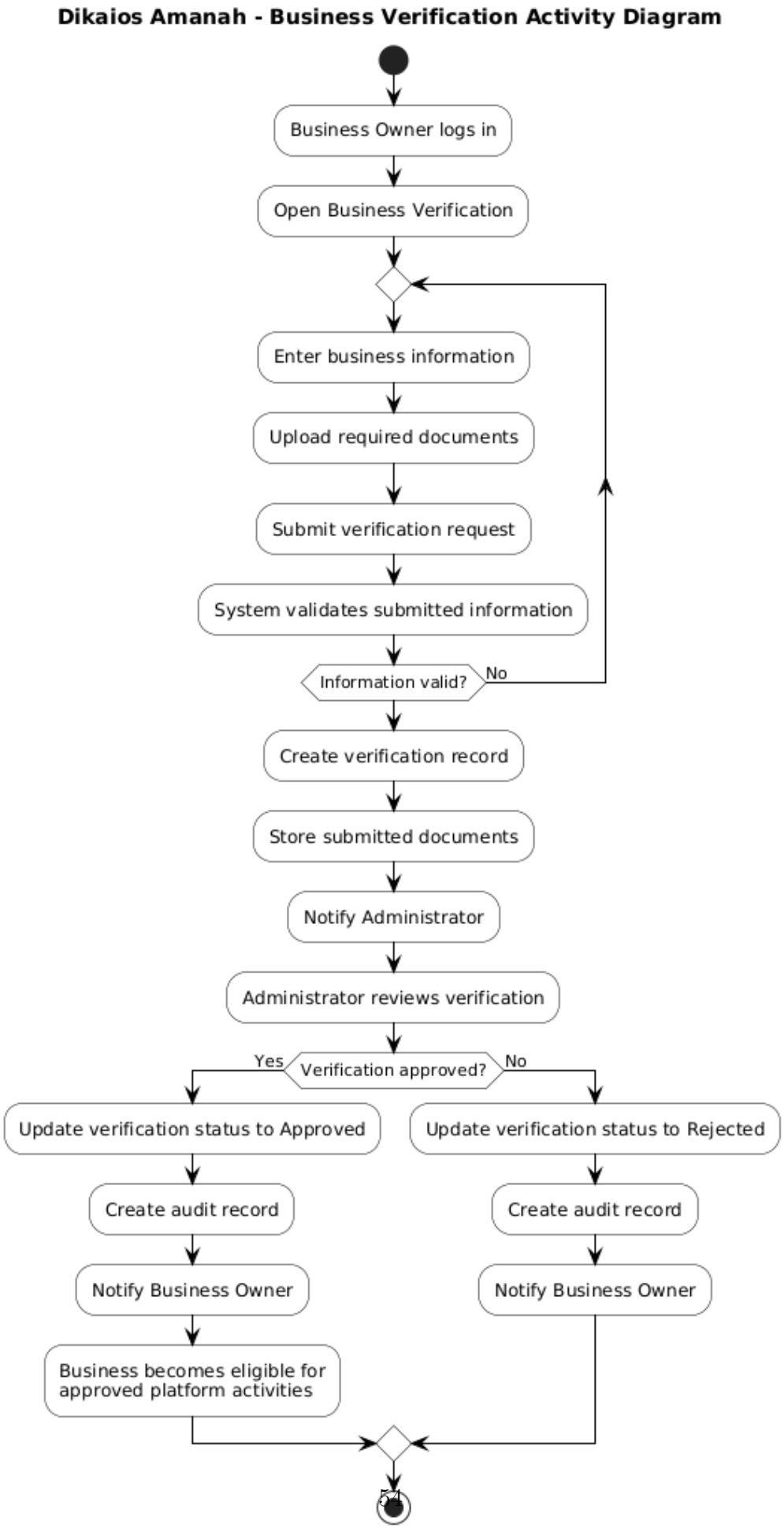


Figure 5: Business Verification Activity Diagram

7.10 System Class Structure

The class diagram presents the principal entities involved in the Dikaios Amanah system, including users, business owners, administrators, businesses, campaigns, investments, verifications, documents, profit declarations, notifications, and audit records.

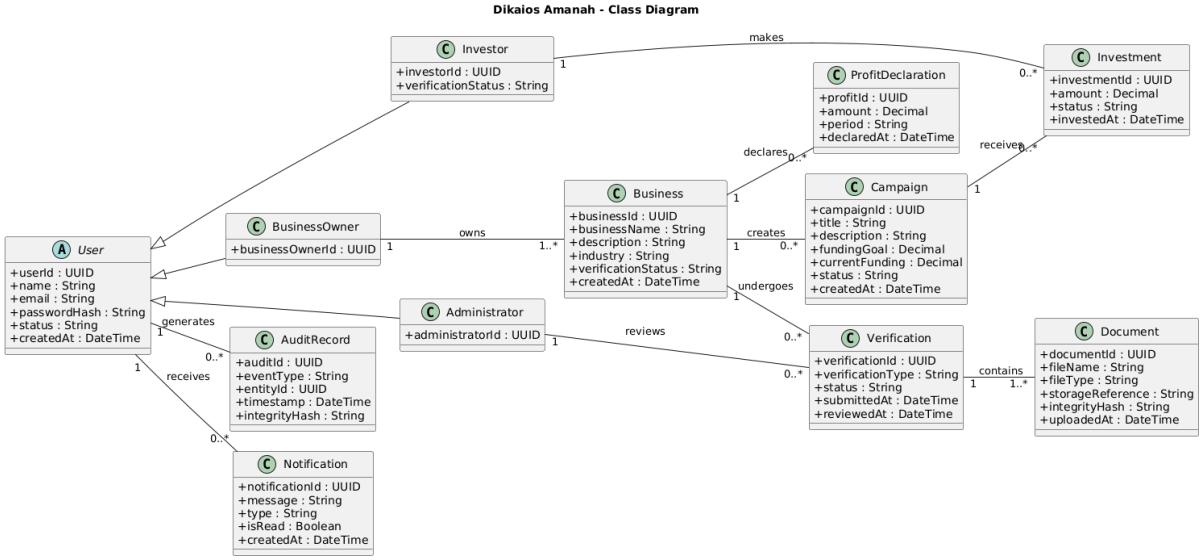


Figure 6: Dikaios Amanah Class Diagram

7.11 System Component Structure

The component diagram presents the major software components of the proposed Dikaios Amanah architecture and their relationships with the application database and document storage.

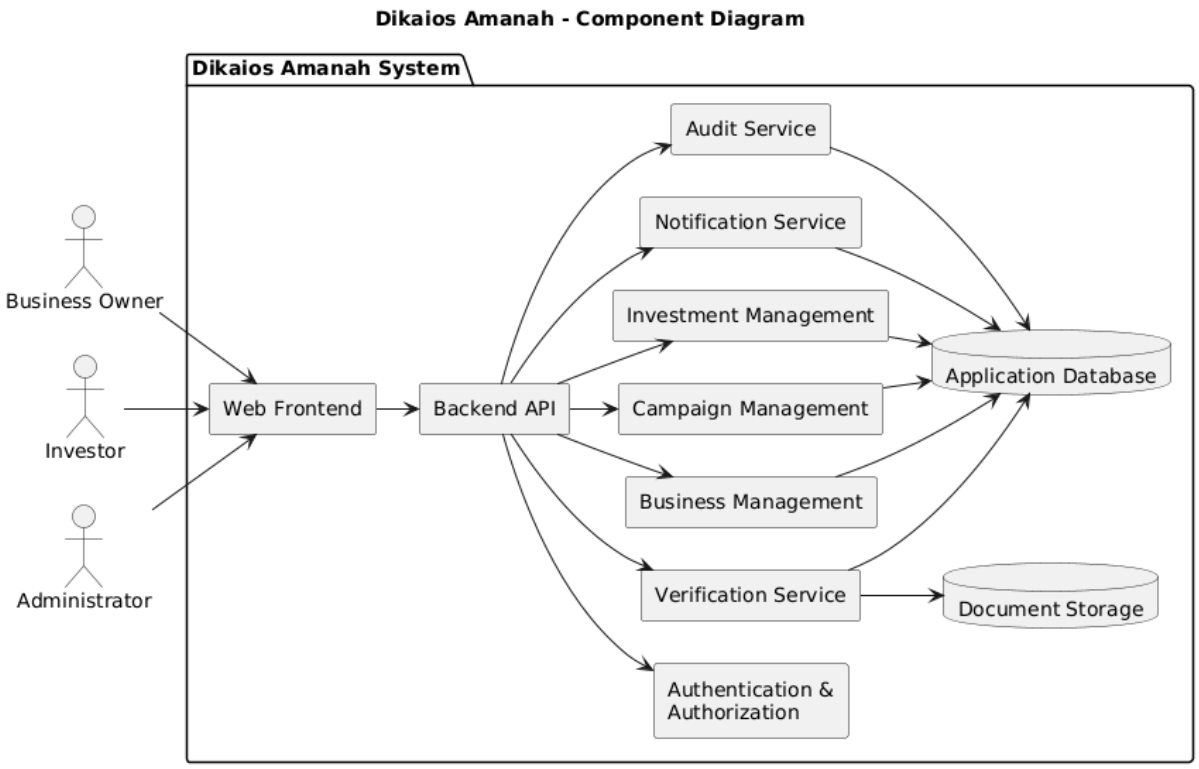


Figure 7: Dikaios Amanah Component Diagram

7.12 Problem Cause Analysis

The Fishbone (Ishikawa) Diagram identifies the major causes contributing to the central problem of lack of trust and transparency in small-business investment.

The identified cause categories include Information, Verification, Transparency, Security, Process, and Investor Awareness.

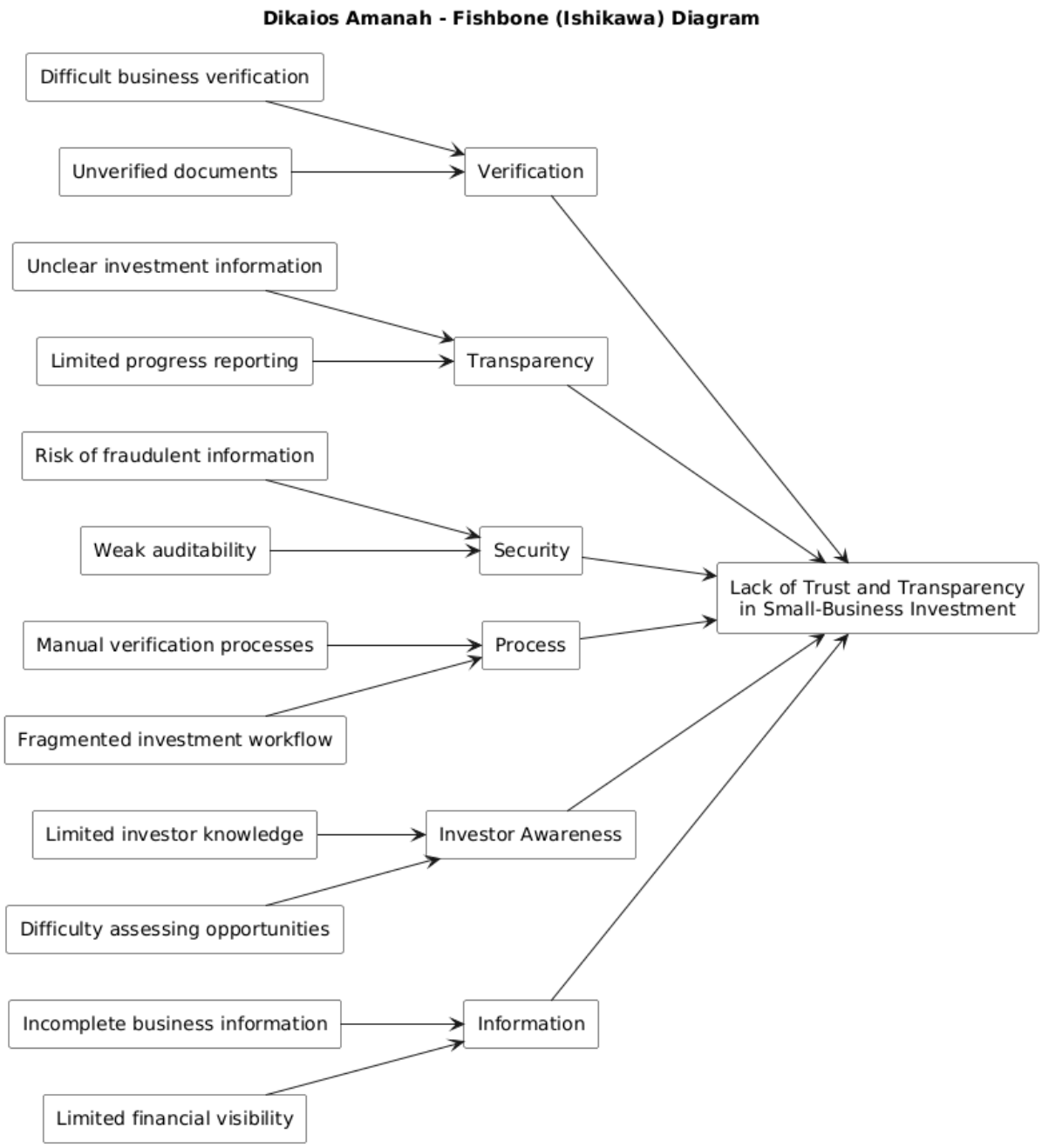


Figure 8: Dikaios Amanah Fishbone (Ishikawa) Diagram

7.13 Relationship to System Requirements

The identified use cases provide the functional foundation for subsequent requirements specification and system design. They identify the primary interactions that must be supported by the system and provide a basis for deriving functional requirements, database entities, system workflows, and software components.

The proposed system scope includes user authentication, business verification, secure

document management, investment campaign creation, campaign browsing, investment tracking, portfolio management, profit declaration, notifications, activity logging, and the Permissioned Trust and Audit Layer.

8. Data Requirements

8.1 Data Requirements

The Dikaios Amanah system requires structured and reliable data to support user management, business verification, investment campaign management, investment participation, profit reporting, notifications, and system auditing. The system shall store and manage the following major categories of data.

8.1.1 User Data

The system shall maintain information required to identify and manage registered users. User data shall include:

- Unique user identifier
- Full name
- Email address
- Authentication credentials or securely stored password hash
- User role
- Account status
- Account creation date and time

The system shall distinguish between the primary user roles of Business Owner, Investor, and Administrator.

8.1.2 Business Data

The system shall maintain information about businesses registered on the platform. Business data shall include:

- Unique business identifier
- Business name
- Business description
- Industry or business category
- Business ownership information
- Verification status
- Business registration information where applicable
- Creation and update timestamps

8.1.3 Verification Data

The system shall maintain information associated with business and user verification processes. Verification data shall include:

- Unique verification identifier
- Verification type
- Related business or user
- Verification status
- Submission date and time
- Review date and time
- Reviewing administrator
- Verification remarks where applicable

The system shall preserve the status of verification requests so that authorized users can determine whether a verification request is pending, approved, or rejected.

8.1.4 Document Data

The system shall maintain metadata associated with documents submitted during verification. Document data shall include:

- Unique document identifier
- Related verification record
- File name
- File type
- Storage reference
- Document integrity information where applicable
- Upload date and time

The system shall restrict access to verification documents according to the user's authorization level.

8.1.5 Investment Campaign Data

The system shall maintain information about investment campaigns created by verified businesses. Campaign data shall include:

- Unique campaign identifier
- Related business
- Campaign title
- Campaign description
- Funding goal
- Current funding amount
- Campaign status
- Creation date and time

- Approval information where applicable

Only campaigns satisfying the required verification and approval conditions shall be made available for participation.

8.1.6 Investment Data

The system shall maintain records of investment activities performed by investors. Investment data shall include:

- Unique investment identifier
- Related investor
- Related campaign
- Investment amount
- Investment status
- Investment date and time

For the scope of the academic project, investment activities represent the system's simulated investment functionality rather than real-world financial transactions.

8.1.7 Profit Declaration Data

The system shall maintain information submitted by businesses regarding their declared profits. Profit declaration data shall include:

- Unique profit declaration identifier
- Related business
- Declared profit amount
- Reporting period
- Declaration date and time
- Relevant status or review information where applicable

8.1.8 Audit Data

The system shall maintain audit records for significant system activities to support transparency and accountability. Audit data shall include:

- Unique audit record identifier
- User associated with the activity
- Event type
- Related entity identifier where applicable
- Date and time of the event
- Integrity information where applicable

Audit records shall be protected from unauthorized modification or deletion.

8.1.9 Notification Data

The system shall maintain notification information for communicating important system events to users. Notification data shall include:

- Unique notification identifier
- Recipient user
- Notification type
- Notification message
- Read or unread status
- Creation date and time

8.1.10 Data Integrity and Security Requirements

The system shall maintain the integrity and confidentiality of stored data. Sensitive information shall not be stored in plaintext where secure storage mechanisms are required.

The system shall enforce appropriate access control so that users can only access data permitted by their roles and responsibilities.

The system shall validate data submitted by users before storing it. Required fields shall be checked for completeness, and invalid or inconsistent data shall be rejected.

The system shall maintain appropriate timestamps for important records to support traceability and auditing.

8.1.11 Data Retention and Backup

Important system records shall be retained for the duration required by the project's operational and academic scope. The system design shall allow appropriate backup and recovery mechanisms to be introduced for persistent data.

As this project is developed as a Software Engineering Lab project, production-level financial data retention and regulatory requirements are outside the current implementation scope.

9. System Constraints and Limitations

9.1 System Constraints and Limitations

The development and operation of the Dikaios Amanah system are subject to several technical, operational, security, and project-level constraints. These constraints define the boundaries of the proposed system and its implementation within the scope of the Software Engineering Lab project.

9.1.1 Project Scope Constraint

Dikaios Amanah is developed as an academic Software Engineering Lab project. Therefore, the system is intended to demonstrate the proposed software architecture, functionality, security concepts, and workflow rather than operate as a production financial platform.

Production-scale deployment, commercial operation, and regulatory certification are outside the current project scope.

9.1.2 Financial Transaction Limitation

The investment functionality implemented within the project represents simulated investment activities. The system does not process real financial transactions, transfer real funds, or provide actual investment services.

Any financial values displayed by the system are intended for demonstration and testing purposes.

9.1.3 Regulatory and Legal Constraints

A production implementation of a platform involving investment activities would be subject to applicable financial, business, data-protection, and other regulatory requirements.

Obtaining licenses, regulatory approvals, legal certifications, and formal authorization to operate a financial service are outside the scope of this academic project.

9.1.4 Third-Party Service Dependency

The system may depend on external services for hosting, cloud infrastructure, file storage, authentication, email or notification delivery, and other supporting functionality.

Availability, pricing, service limits, and changes to third-party services may affect system operation.

9.1.5 Internet and Infrastructure Dependency

The web-based system requires network connectivity for communication between the client application, backend services, database, and supporting cloud infrastructure.

Temporary network failures or service outages may prevent users from accessing certain system functionality.

9.1.6 Document Verification Limitation

The system can collect and manage documents submitted for verification, but the academic implementation cannot guarantee that submitted documents are legally authentic.

Document review and verification functionality is intended to demonstrate the software workflow and access-control mechanisms rather than replace formal legal or governmental verification procedures.

9.1.7 Security Constraints

The system shall implement appropriate authentication, authorization, input validation, secure credential handling, and audit mechanisms within the project's technical scope.

However, the academic implementation cannot guarantee complete protection against every possible cybersecurity threat. Production-level penetration testing, security certification, continuous security monitoring, and advanced threat detection are outside the current scope.

9.1.8 Scalability Constraint

The initial implementation is designed for demonstration and academic evaluation rather than large-scale commercial usage.

The system may therefore require additional infrastructure, optimization, caching, database scaling, and load-balancing mechanisms before supporting a large number of concurrent users.

9.1.9 Data Availability Constraint

The quality of system demonstrations and testing depends on the availability of suitable test data.

The project may use controlled, synthetic, or sample data for demonstration purposes. Real confidential business, financial, or personal information shall not be required for the academic implementation.

9.1.10 Development Resource Constraint

The project is developed within the time, hardware, software, and resource limitations of an academic Software Engineering Lab project.

These limitations may restrict the number of advanced features that can be implemented during the project period.

9.1.11 Deployment Constraint

The final system may be deployed using cloud-based infrastructure appropriate for the project's requirements. However, the deployment configuration is intended for demonstra-

tion and evaluation and may not provide the redundancy, monitoring, disaster recovery, and operational support expected from a production financial platform.

10. Future Enhancements

10.1 Future Enhancements

The current Dikaios Amanah system is designed as an academic MVP. Several features and improvements can be considered for future versions if the system is extended beyond the scope of the Software Engineering Lab project.

10.1.1 Real Financial Transaction Integration

A future production-oriented version could integrate with authorized financial institutions or payment service providers to support real financial transactions.

Such functionality would require appropriate regulatory approval, security controls, financial compliance mechanisms, and integration with authorized payment infrastructure.

10.1.2 Advanced Business Verification

The verification system could be extended to integrate with official business registries and authorized verification services.

Automated verification could reduce manual review requirements and improve the reliability and speed of the verification process.

10.1.3 Advanced Risk Assessment

Future versions could introduce more sophisticated risk assessment mechanisms using historical business performance, financial indicators, market information, and other relevant factors.

Machine learning techniques could also be investigated to assist with risk classification and decision support.

10.1.4 Enhanced Fraud Detection

The system could be extended with automated fraud detection mechanisms to identify suspicious accounts, unusual investment patterns, document inconsistencies, and other potentially fraudulent activities.

10.1.5 Mobile Application

A dedicated mobile application could be developed to provide Business Owners, Investors, and Administrators with convenient access to the platform through mobile devices.

10.1.6 Advanced Notification System

Future versions could support multiple notification channels, including email, SMS, and push notifications, for important verification, campaign, investment, and administrative events.

10.1.7 Improved Scalability

The system could be expanded to support larger numbers of users and higher transaction volumes through additional infrastructure, database optimization, caching, load balancing, and distributed services.

10.1.8 Enhanced Analytics and Reporting

Advanced dashboards and reporting capabilities could be introduced to provide Business Owners, Investors, and Administrators with more detailed insights into business performance, campaign progress, investment activity, and platform activity.

10.1.9 Expanded Trust and Audit Infrastructure

The Permissioned Trust and Audit Layer could be expanded to provide stronger tamper-evident record keeping and more comprehensive audit capabilities for critical system events.

10.1.10 Regulatory and Production Readiness

If the system were developed beyond the academic environment, future work would include formal security assessment, regulatory compliance, privacy controls, operational monitoring, disaster recovery, and other requirements necessary for production deployment.

11. Testing Requirements

11.1 Testing Requirements

Testing shall be performed to verify that Dikaios Amanah satisfies its functional and non-functional requirements. The testing process shall focus on the correctness of the major system workflows, role-based access control, data validation, security mechanisms, and system integration.

11.1.1 Unit Testing

Individual software components and functions shall be tested independently to verify that they produce the expected results.

Unit testing should cover important backend services, validation functions, authentication-related functions, and other critical application logic.

11.1.2 Integration Testing

Integration testing shall verify that different system components communicate correctly.

The following integrations shall be considered:

- Frontend and backend API communication.
- Backend and PostgreSQL database communication.
- Backend and document storage.
- Authentication and authorization services.
- Backend and the trust and audit layer.

11.1.3 System Testing

System testing shall evaluate the complete application as an integrated system.

Testing shall cover the major workflows of Business Owners, Investors, and Administrators, including business verification, campaign management, investment simulation, profit declaration, and administrative operations.

11.1.4 Functional Testing

Functional tests shall verify that the implemented system behavior matches the functional requirements.

Representative functional test cases shall include:

- User registration and login.
- Role-based access control.
- Business profile creation and management.
- Business verification submission.
- Document upload and management.
- Campaign creation and approval.
- Campaign browsing.
- Simulated investment.
- Portfolio and investment history.
- Profit declaration.
- Administrative review and monitoring.
- Audit record generation.

11.1.5 Security Testing

Security testing shall verify that protected system resources cannot be accessed by unauthorized users.

Security testing shall include:

- Authentication testing.
- Authorization and role testing.
- Invalid credential testing.
- Protected endpoint testing.
- Input validation testing.
- Unauthorized document access testing.
- Session or token validation testing.

11.1.6 Usability Testing

Usability testing shall evaluate whether users can understand and complete the major system workflows using the provided interface.

The primary user roles shall be considered during usability testing, including Business Owners, Investors, and Administrators.

11.1.7 Performance Testing

Basic performance testing shall be performed to evaluate the system's response under representative academic testing conditions.

Testing may include:

- API response time.
- Page loading behavior.
- Database query performance.
- Response behavior under multiple simulated requests.

Production-scale load and stress testing are outside the current academic project scope.

11.1.8 Negative Testing

The system shall be tested with invalid, incomplete, or unexpected input to verify that errors are handled appropriately.

Examples include:

- Invalid login credentials.
- Missing required fields.
- Invalid document types.
- Invalid investment amounts.
- Unauthorized operations.
- Invalid campaign information.

11.1.9 User Acceptance Testing

User Acceptance Testing (UAT) shall verify that the system satisfies the expected workflows from the perspective of its intended users.

Representative scenarios shall be executed for the Business Owner, Investor, and Administrator roles.

11.1.10 Test Documentation

Test cases shall document the following information where applicable:

- Test case identifier.
- Related requirement.
- Test objective.
- Preconditions.
- Test input.

- Expected result.
- Actual result.
- Test status.

Test results shall be used to identify defects and verify that corrective changes satisfy the corresponding requirements.

12. Requirements Traceability

12.1 Requirements Traceability

Requirements traceability establishes a relationship between the identified system requirements, their implementation, and the testing activities used to verify them.

The traceability process helps ensure that each major requirement is addressed by the system design and can be verified through appropriate testing.

12.1.1 Traceability Approach

Each functional requirement shall have a unique identifier. These identifiers shall be used consistently throughout the system design, implementation, and testing documentation.

The main traceability relationship is:

Requirement → Design → Implementation → Test Case

This approach allows the development team to determine whether a requirement has been implemented and whether sufficient testing has been performed.

12.1.2 Functional Requirement Traceability

The following table provides a high-level mapping between the major functional requirements and their corresponding system areas and testing activities.

Requirement	System Area	Design Element	Verification
FR-AUTH	Authentication and account management	Authentication module	Authentication tests
FR-BO	Business Owner functionality	Business management and campaign modules	Business Owner functional tests
FR-INV	Investor functionality	Investment and portfolio modules	Investor functional tests
FR-ADM	Administrator functionality	Administration module	Administrator functional tests
FR-VER	Business verification	Verification and document modules	Verification test cases
FR-CAM	Campaign management	Campaign module	Campaign functional tests
FR-INVEST	Investment functionality	Investment module	Investment test cases
FR-PROFIT	Profit declaration	Profit management module	Profit declaration tests
FR-AUDIT	Audit and traceability	Audit and Trust Layer	Audit verification tests

Table 2: High-Level Functional Requirement Traceability

12.1.3 Non-Functional Requirement Traceability

Non-functional requirements shall also be considered during system design and testing.

Requirement	Related System Area	Verification Method
Security	Authentication, authorization, input validation	Security testing
Performance	Backend, database, and frontend	Performance testing
Usability	User interface	Usability testing
Reliability	Application and database services	System and integration testing
Maintainability	Software architecture and modular components	Code and architecture review

Table 3: Non-Functional Requirement Traceability

12.1.4 Traceability During Development

The traceability information shall be maintained throughout the development process. Changes to important requirements should be reflected in the relevant design components, implementation tasks, and test cases.

This approach will help reduce the possibility of implementing system features that cannot be traced back to an identified requirement and will support systematic verification of the final system.

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